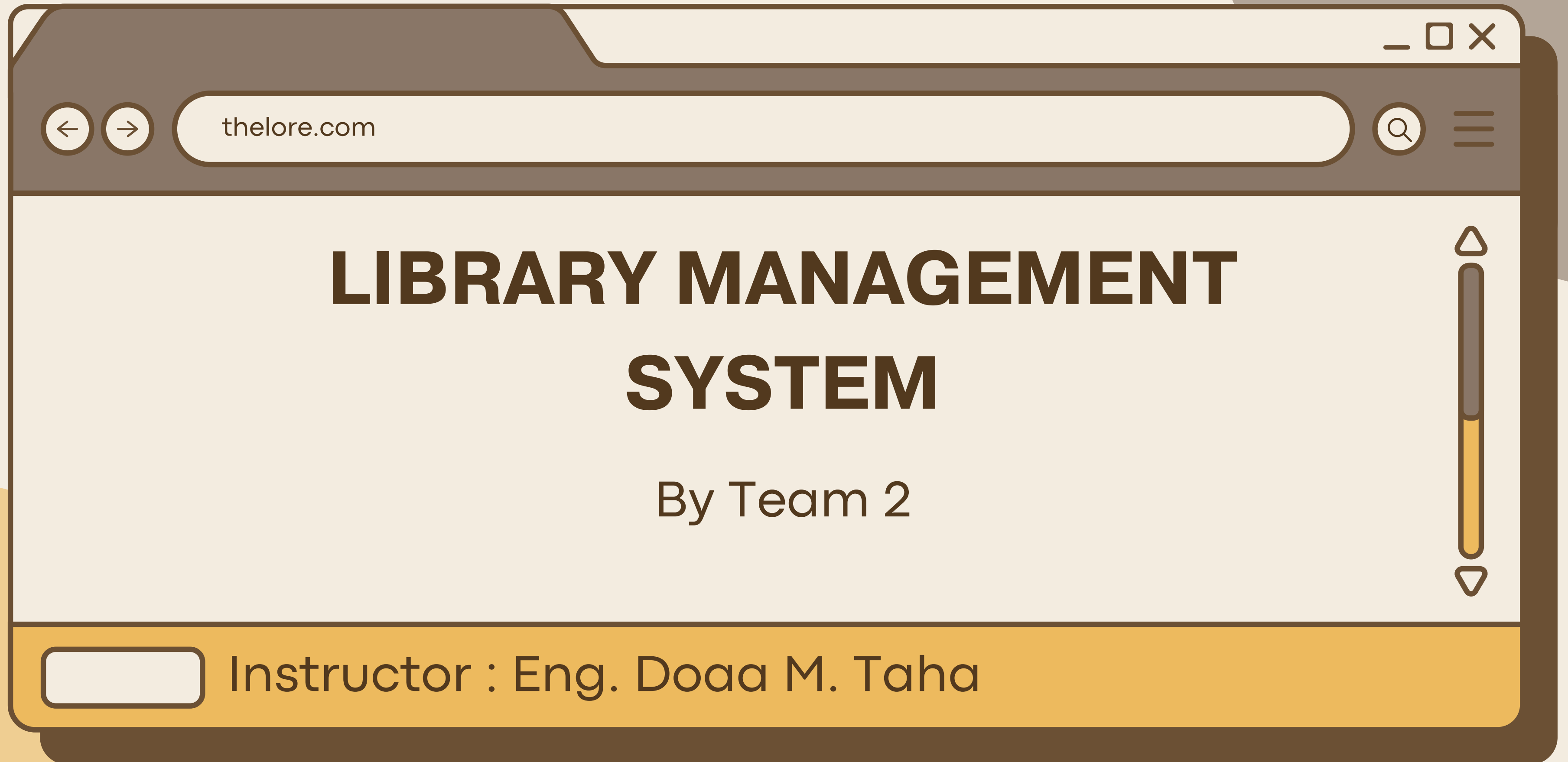
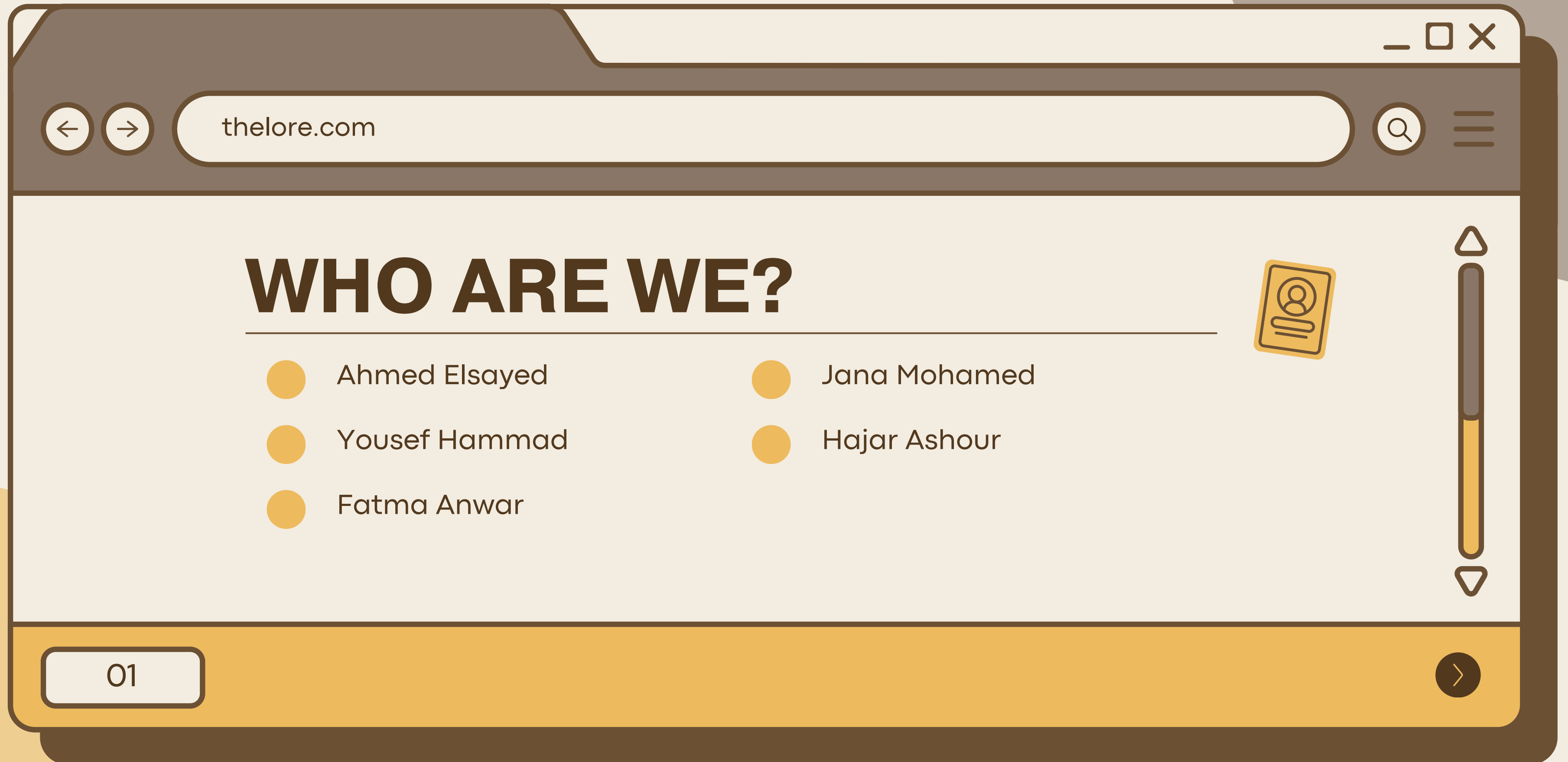




LIBRARY MANAGEMENT SYSTEM

By Team 2

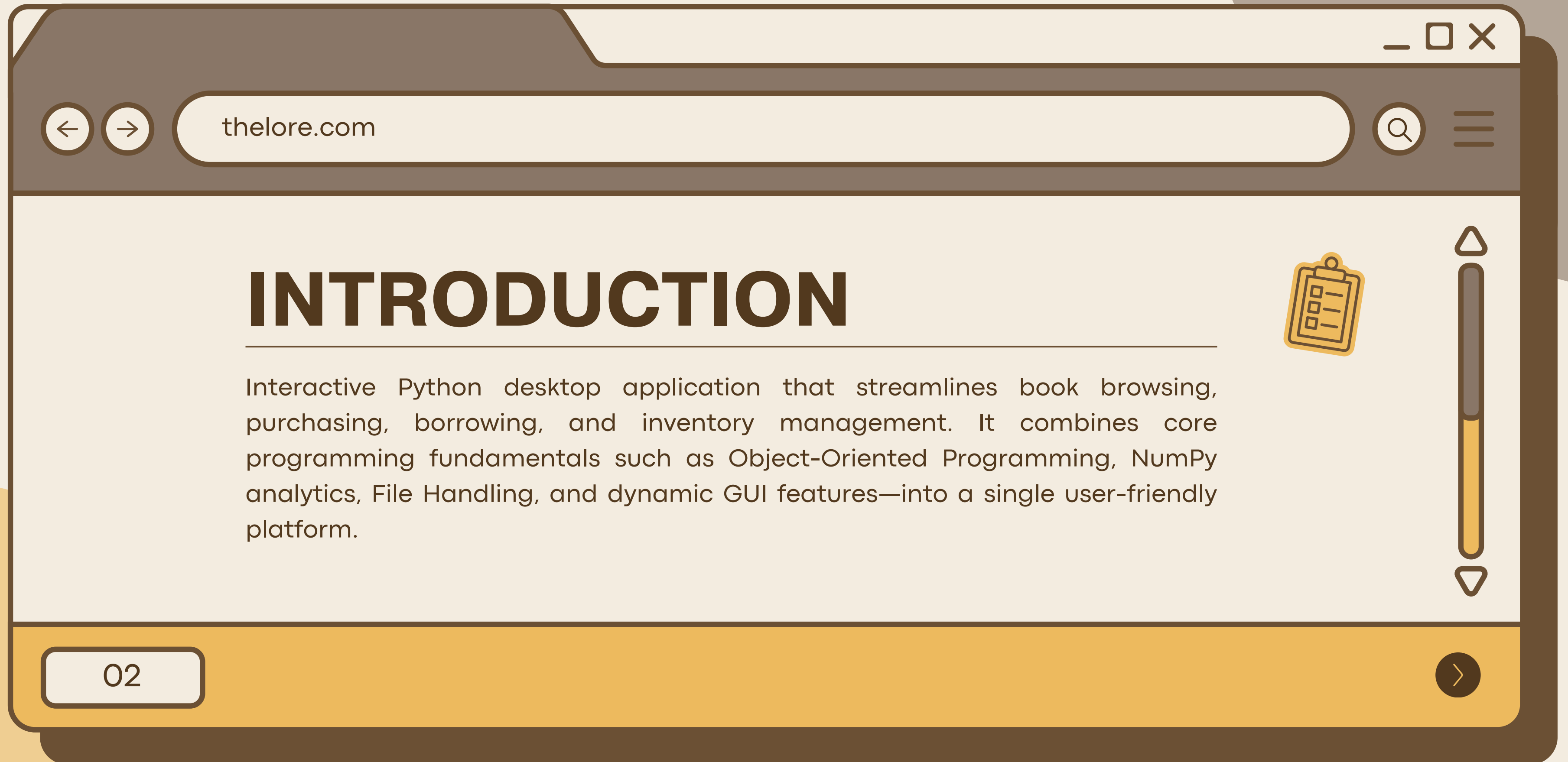
Instructor : Eng. Dood M. Taha



WHO ARE WE?

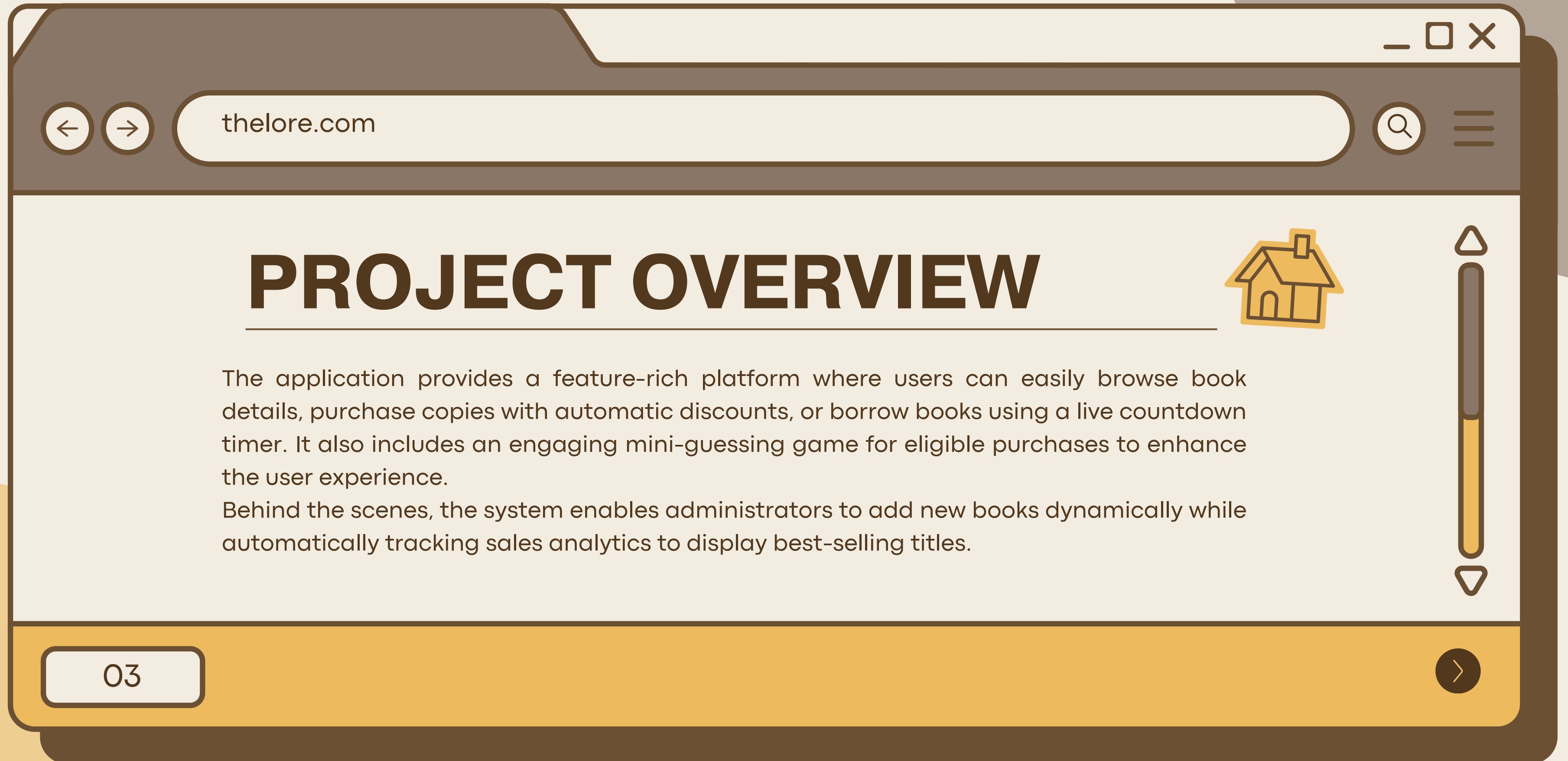
- Ahmed Elsayed
- Yousef Hammad
- Fatma Anwar
- Jana Mohamed
- Hajar Ashour





INTRODUCTION

Interactive Python desktop application that streamlines book browsing, purchasing, borrowing, and inventory management. It combines core programming fundamentals such as Object-Oriented Programming, NumPy analytics, File Handling, and dynamic GUI features—into a single user-friendly platform.



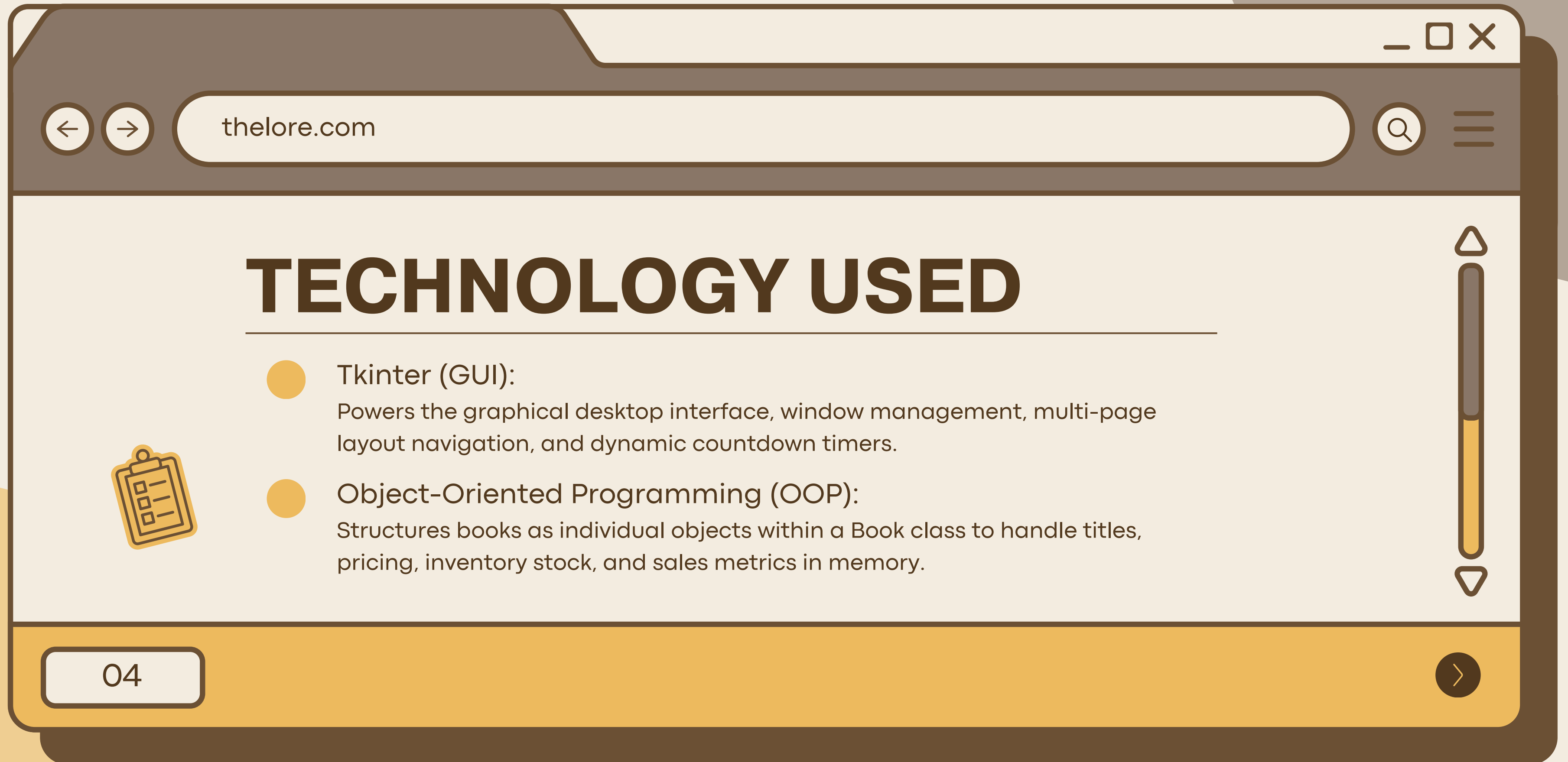
thelore.com

PROJECT OVERVIEW



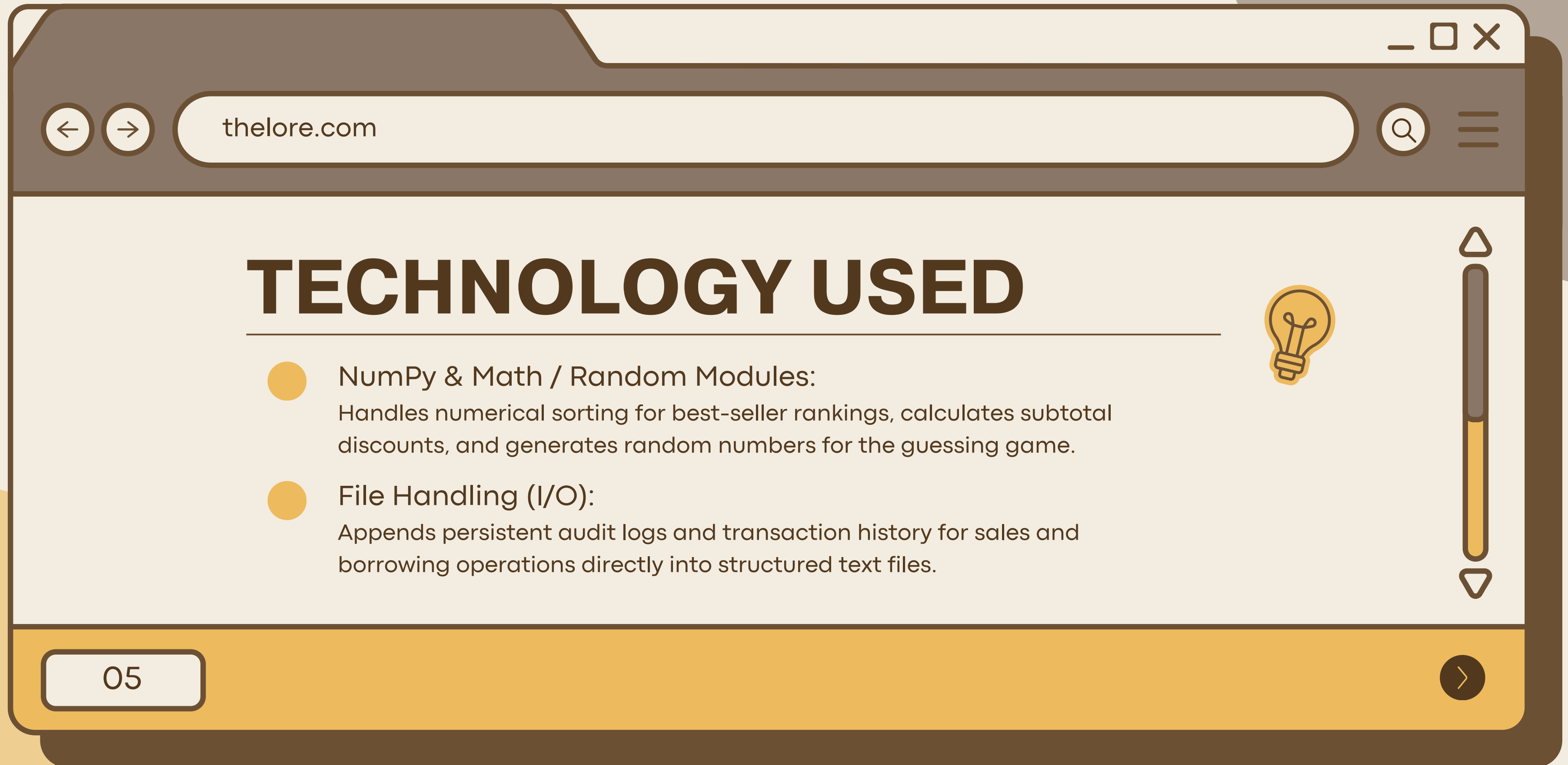
The application provides a feature-rich platform where users can easily browse book details, purchase copies with automatic discounts, or borrow books using a live countdown timer. It also includes an engaging mini-guessing game for eligible purchases to enhance the user experience.

Behind the scenes, the system enables administrators to add new books dynamically while automatically tracking sales analytics to display best-selling titles.



TECHNOLOGY USED

- Tkinter (GUI):
Powers the graphical desktop interface, window management, multi-page layout navigation, and dynamic countdown timers.
- Object-Oriented Programming (OOP):
Structures books as individual objects within a Book class to handle titles, pricing, inventory stock, and sales metrics in memory.

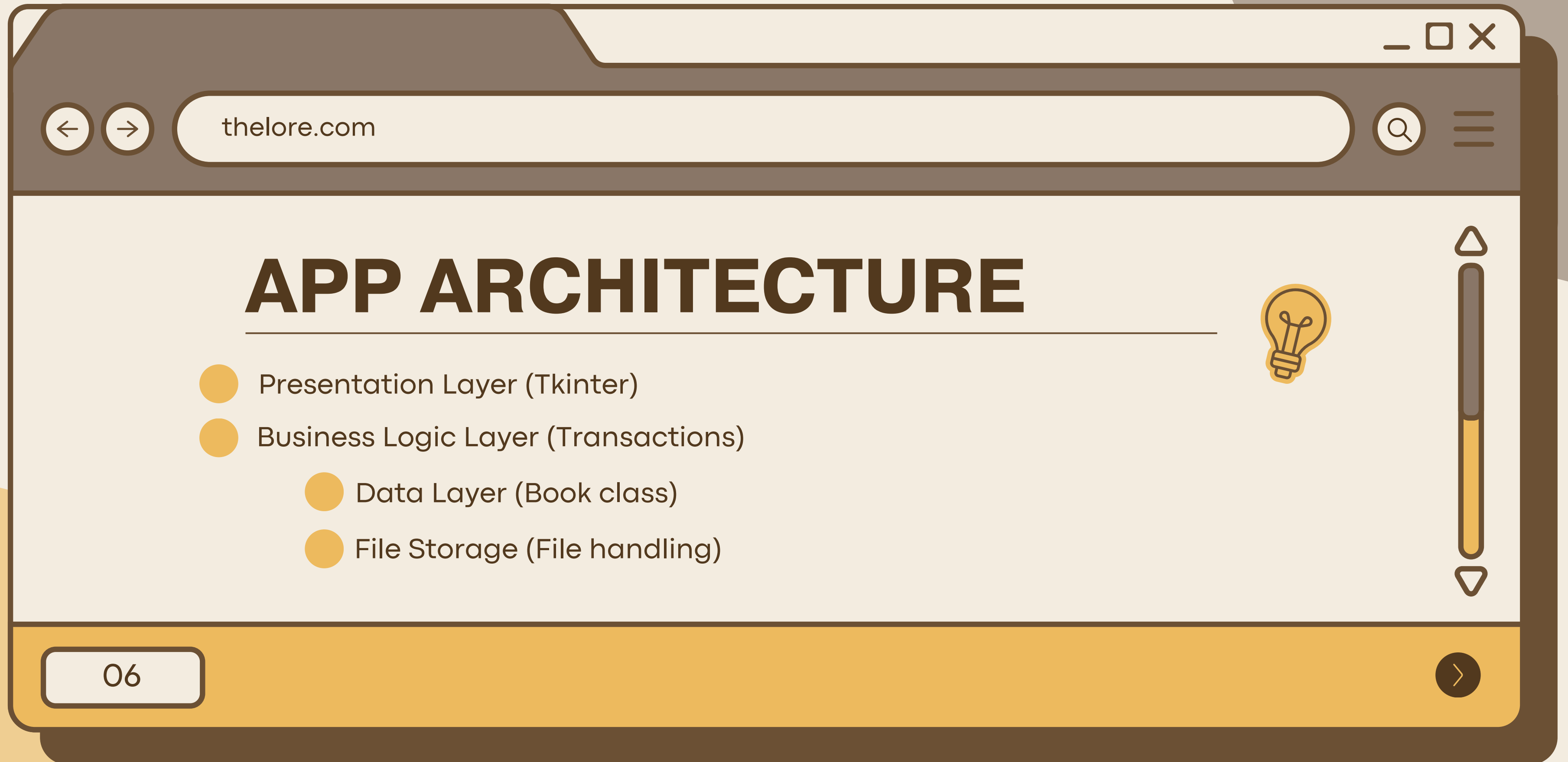


thelore.com

TECHNOLOGY USED

- NumPy & Math / Random Modules:
Handles numerical sorting for best-seller rankings, calculates subtotal discounts, and generates random numbers for the guessing game.
- File Handling (I/O):
Appends persistent audit logs and transaction history for sales and borrowing operations directly into structured text files.

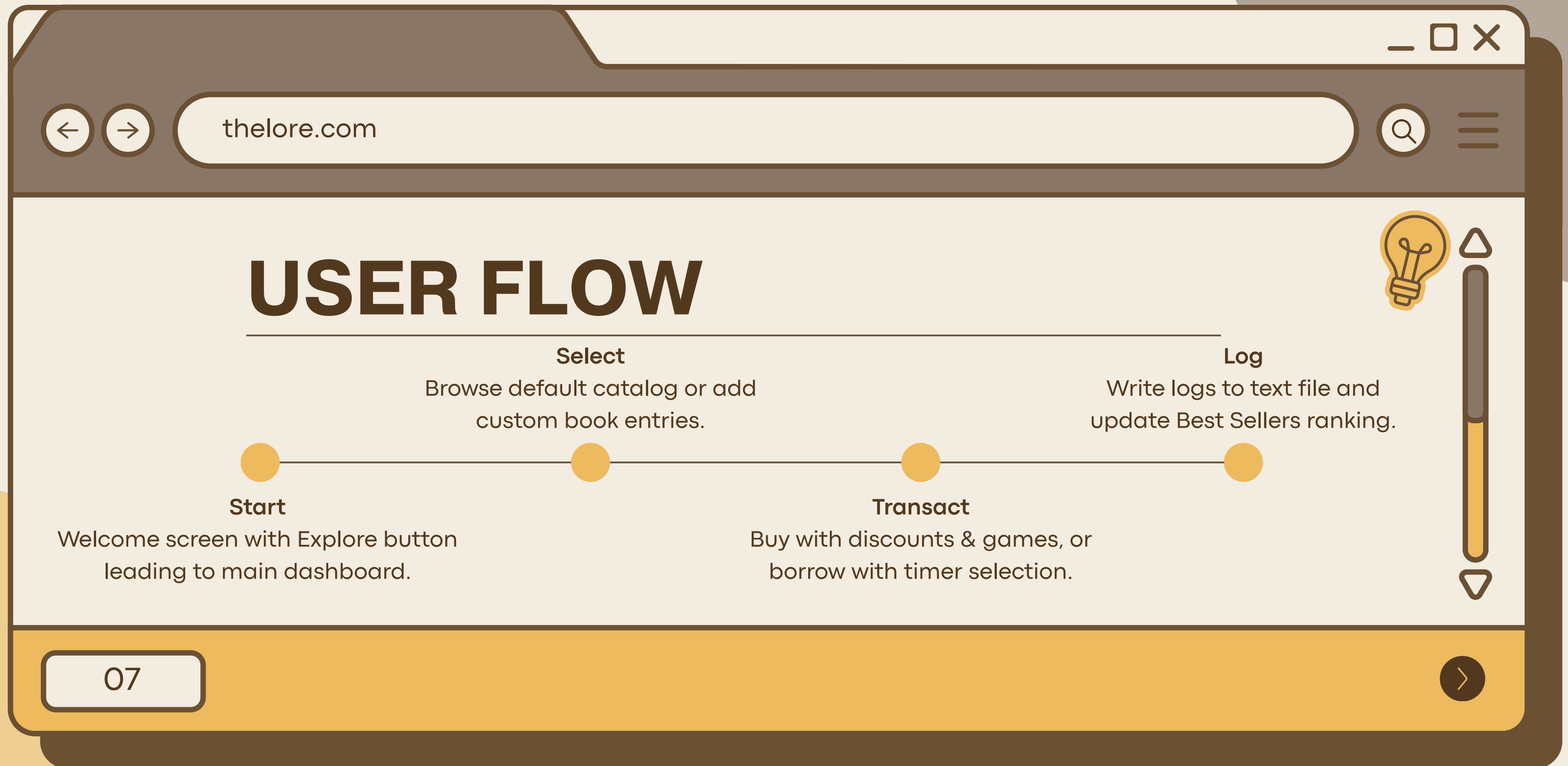


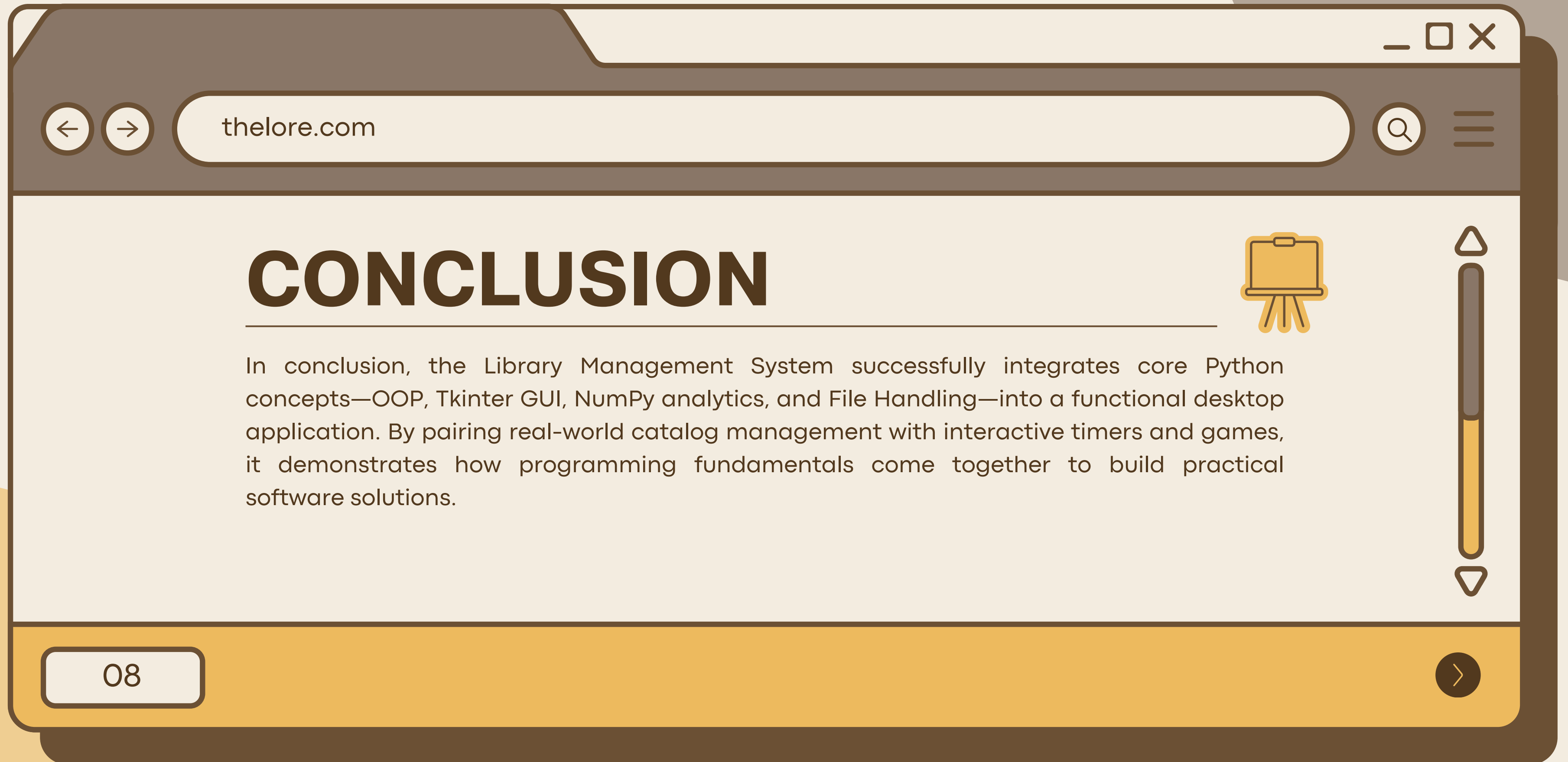


APP ARCHITECTURE

- Presentation Layer (Tkinter)
- Business Logic Layer (Transactions)
 - Data Layer (Book class)
 - File Storage (File handling)







CONCLUSION



In conclusion, the Library Management System successfully integrates core Python concepts—OOP, Tkinter GUI, NumPy analytics, and File Handling—into a functional desktop application. By pairing real-world catalog management with interactive timers and games, it demonstrates how programming fundamentals come together to build practical software solutions.



